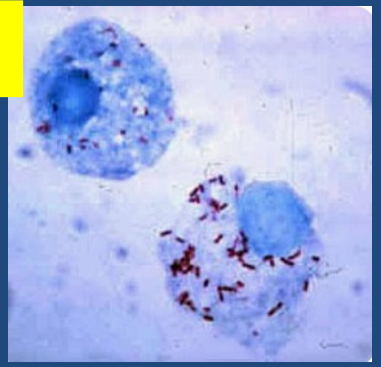
Bacteria and Other Microorganisms

human infectious diseases belong to five major groups of organisms:

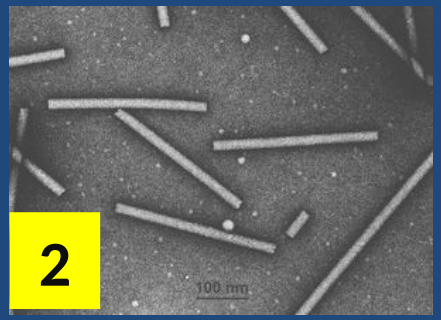
- Bacteria.
- Fungi.
- Parasites (Protozoa and Helminths).
- Viruses

Kingdom	Pathogenic Microorganisms	Type of Cells
Animal	Helminths (worms)	Eukaryotic
Protists	Protozoa	Eukaryotic
Fungi	Fungi (yeasts and molds)	Eukaryotic
Prokaryote	Bacteria	Prokaryotic
	Viruses	Noncellular

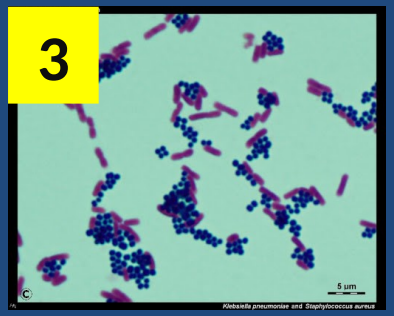
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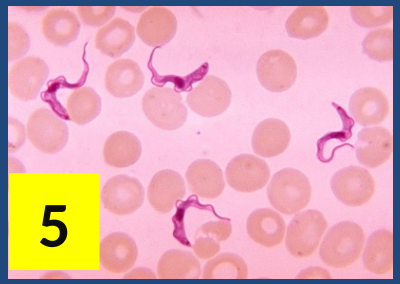
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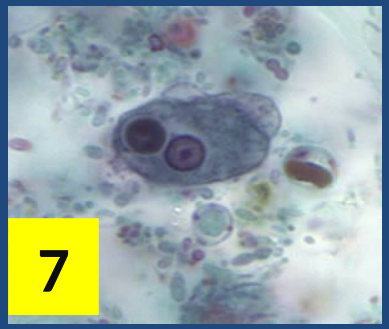
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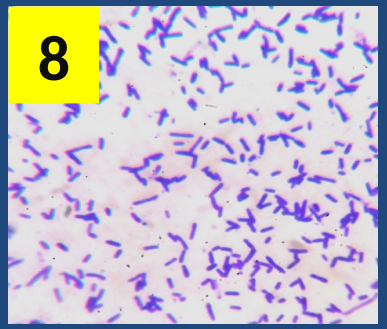
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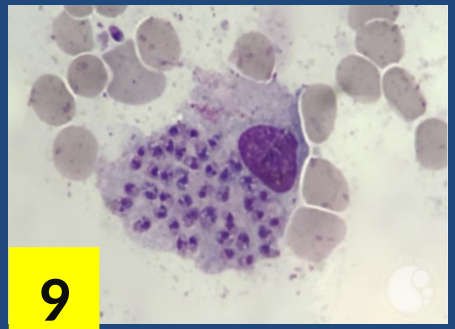
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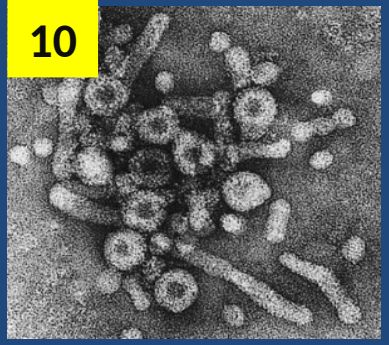
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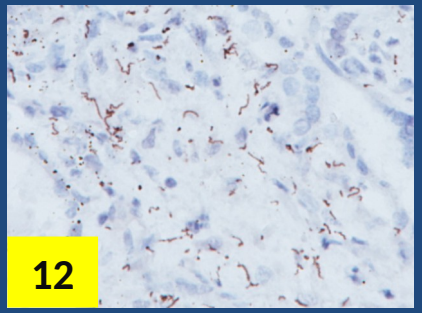
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Comparison of Medically Important Organisms

Characteristic	Viruses	Bacteria	Fungi	Protozoa and Helminths
Cells	No	Yes	Yes	Yes
Approximate diameter (µm)[1]	0.02–0.2	1–5	3–10 (yeasts)	15–25 (trophozoites)
Nucleic acid	Either DNA or RNA	Both DNA and RNA	Both DNA and RNA	Both DNA and RNA
Type of nucleus	None	Prokaryotic	Eukaryotic	Eukaryotic
Ribosomes	Absent	70S	80S	80S
Mitochondria	Absent	Absent	Present	Present
Nature of outer surface	Protein capsid and lipoprotein envelope	Rigid wall containing peptidoglycan	Rigid wall containing chitin	Flexible membrane
Motility	None	Some	None	Most
Method of replication	Not binary fission	Binary fission	Budding or mitosis[2]	Mitosis[3]

Three main differential criteria based on:

1- Structure:

- Cells have a nucleus or nucleoid which contains DNA; this is surrounded by cytoplasm, within which proteins are synthesized and energy is generated.
- Viruses have genetic material (either DNA or RNA) but no cytoplasm, and so they depend on host cells to provide the machinery for protein synthesis and energy generation.

2- Method of replication:

- Cells replicate either by binary fission or by mitosis, during which one parent cell divides to make two progeny cells while retaining its cellular structure.
- Prokaryotic cells (e.g., bacteria) replicate by binary fission, whereas eukaryotic cells replicate by mitosis.
- viruses disassemble, produce many copies of their nucleic acid and protein, and then reassemble into multiple progeny viruses.

3- Nature of the nucleic acid:

- Cells contain both DNA and RNA, whereas viruses contain either DNA or RNA, but not both.

EUKARYOTES and PROKARYOTES

- The eukaryotic cell has a true nucleus with multiple chromosomes surrounded by a nuclear membrane and uses a mitotic apparatus to ensure equal allocation of the chromosomes to progeny cells.

- The nucleoid of a prokaryotic cell consists of a single circular molecule of loosely organized DNA, lacking a nuclear membrane and mitotic apparatus

Characteristic	Prokaryotic Bacterial Cells	Eukaryotic Human Cells
DNA within a nuclear membrane	No	Yes
Mitotic division	No	Yes
DNA associated with histones	No	Yes
Chromosome number	One	More than one
Membrane-bound organelles, such as mitochondria and lysosomes	No	Yes
Size of ribosome	70S	80S
Cell wall containing peptidoglycan	Yes	No

- Eukaryotic cells: contain organelles, such as mitochondria and lysosomes, and larger (80S) ribosomes.

- Prokaryotic cells: contain no organelles and smaller (70S) ribosomes.

- Most prokaryotes have a rigid external cell wall that contains peptidoglycan.

- Eukaryotes, do not contain peptidoglycan. Either they are bound by a flexible cell membrane, or, in the case of fungi, they have a rigid cell wall (composed from chitin).

- The eukaryotic cell membrane contains sterols, whereas no prokaryote, except the wall-less *Mycoplasma*, has sterols in its membranes.

Terminology and Classification

- Bacteria, fungi, protozoa, and helminths are named according to the binomial Linnean system that uses genus and species.
- Viruses have single names (Informal names: spheres, rods,,,).

Examples:
- *Escherichia coli*, *Escherichia* is the genus and *coli* is the species name.
- *Candida albicans* consists of *Candida* as the genus and *albicans* as the species. (Bright white!!!)
- Viruses: poliovirus, measles virus, or rabies virus. Names with two parts: herpes simplex virus.

- The suffix "-ella" is added to the names of some bacterial genera (examples: *Legionella* and *Klebsiella*) to honour scientists.

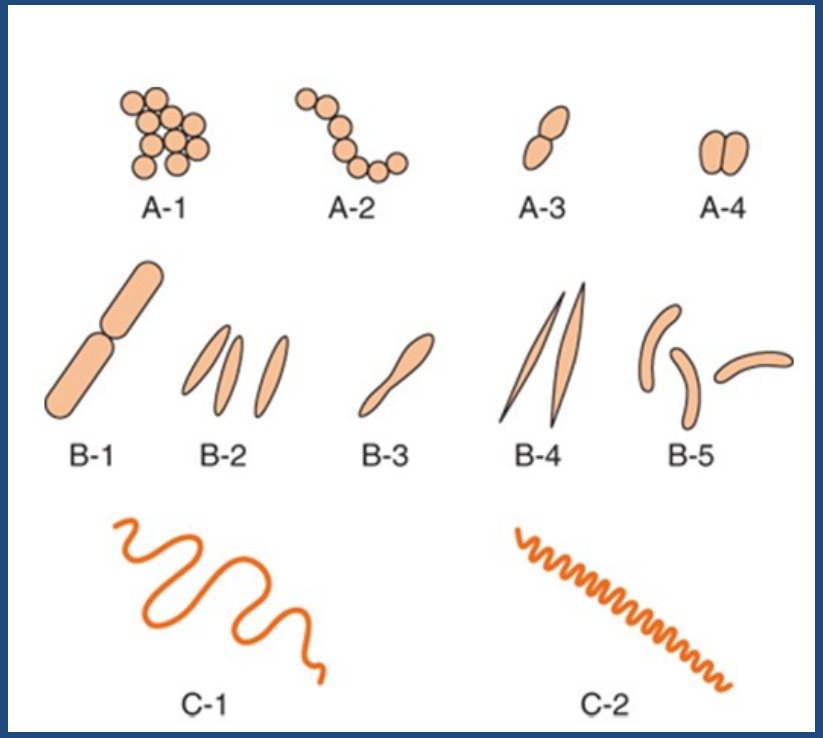
Structure of Bacterial Cells

- Bacteria are classified by shape

1- Cocci, 2- Bacilli, 3- Spirochetes.

- Some bacteria are variable in shape and are said to be pleomorphic (many-shaped).

- The microscopic appearance of a bacterium is one of the most important criteria used in its identification.



A: Cocci in clusters (e.g., *Staphylococcus*; A-1); chains (e.g., *Streptococcus*; A-2); in pairs with pointed ends (e.g., *Streptococcus pneumoniae*; A-3); in pairs with kidney bean shape (e.g., *Neisseria*; A-4). **B:** Rods (bacilli): with square ends (e.g., *Bacillus*; B-1); with rounded ends (e.g., *Salmonella*; B-2); club-shaped (e.g., *Corynebacterium*; B-3); fusiform (e.g., *Fusobacterium*; B-4); comma-shaped (e.g., *Vibrio*; B-5). **C:** Spirochetes: relaxed coil (e.g., *Borrelia*; C-1); tightly coiled (e.g., *Treponema*; C-2).

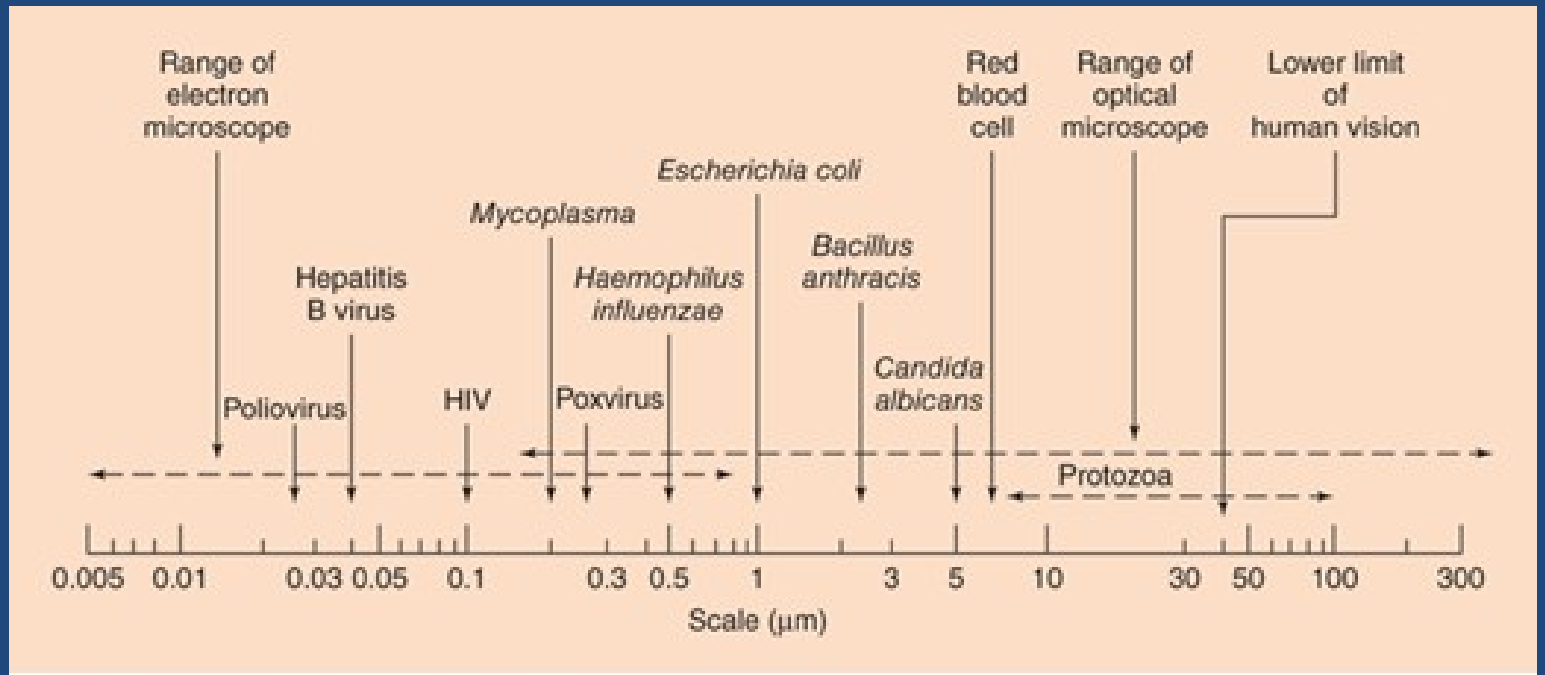
- In addition to their characteristic shapes, the arrangement of bacteria is important.
- Certain cocci occur in pairs (diplococci), some in chains (streptococci), and others in grapelike clusters (staphylococci).
- These arrangements are determined by the orientation and degree of attachment of the bacteria at the time of cell division.
- The arrangement of rods and spirochetes is medically less important.

Bacteria size:

- Range from about 0.2 to 5 µm.

- The smallest bacteria (*Mycoplasma*) are about the same size as the largest viruses (poxviruses).

- The longest bacteria rods are the size of some yeasts and human red blood cells (7 µm).



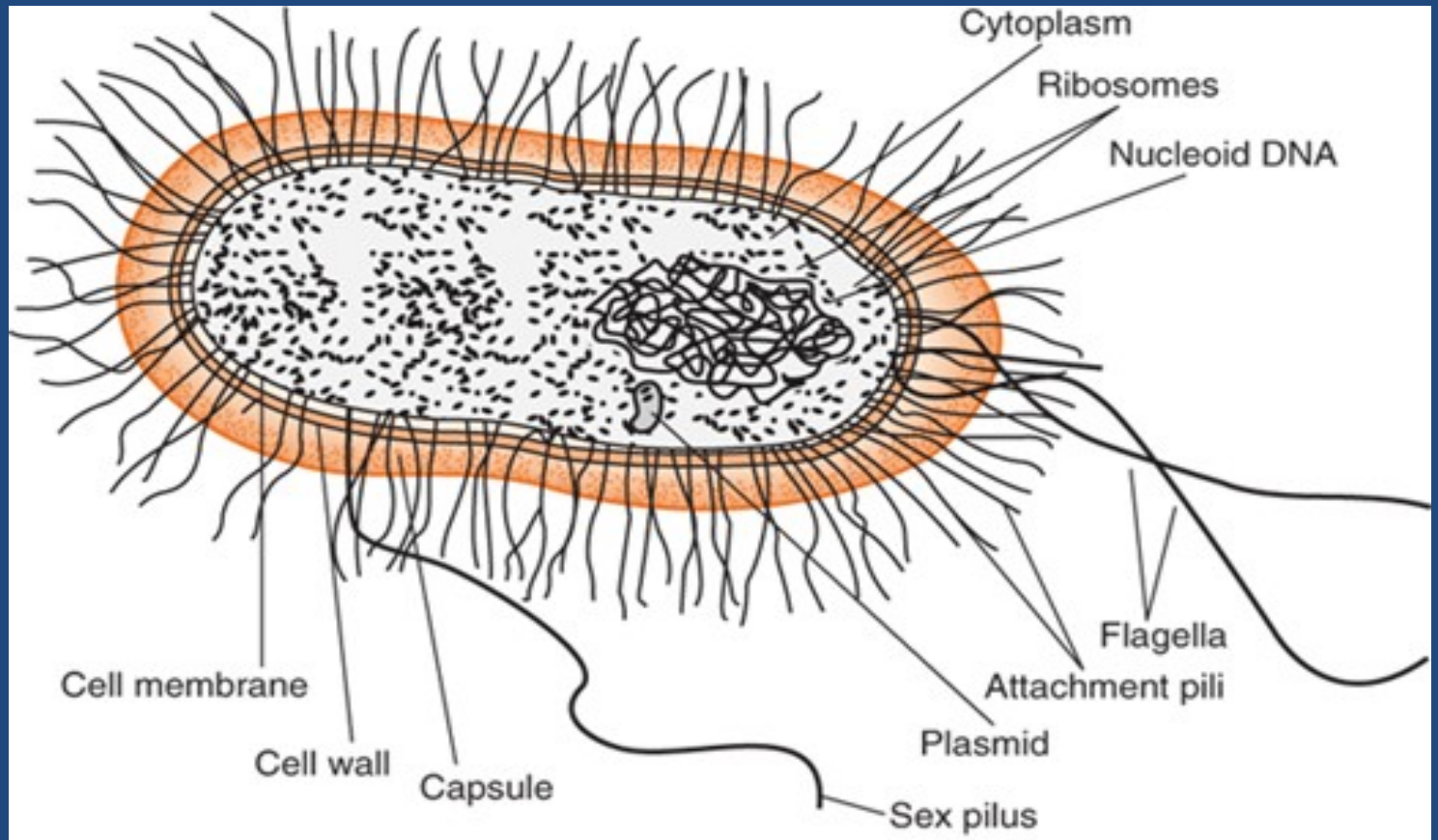
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Prokaryotic cell: Architectural regions

1. **Appendages** (attachments to the cell surface)
   - **Flagella**
   - **Pili (or fimbriae)**

2. **Cell envelope** consisting of
   - **Capsule**
   - **Cell wall**
   - **Plasma membrane**

3. **Cytoplasmic region** that contains
   - **Chromosome** (**DNA**)
   - **Ribosomes** and
   - Various sorts of **inclusions**



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Cell Wall

- Outermost component of bacterial cell.

- *Mycoplasma* species, are bounded by a cell membrane and they don't have cell wall.

- Some bacteria have surface features: such as a capsule, flagella, and pili.

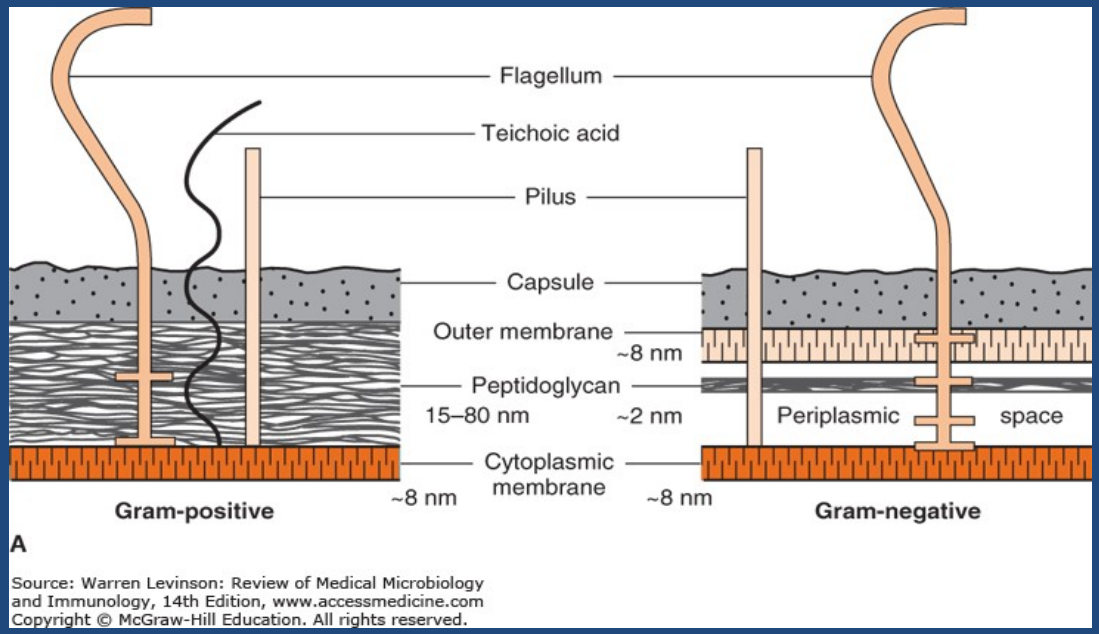
- The cell wall is composed of peptidoglycan.

- The peptidoglycan provides structural support and maintains the characteristic shape of the cell.

- Differences between cell walls of Gram-positive and Gram-negative bacteria will be discussed.

Comparison of Cell Walls of Gram-Positive and Gram-Negative Bacteria

Component	Gram-Positive Cells	Gram-Negative Cells
Peptidoglycan	Thicker; multilayer	Thinner; single layer
Teichoic acids	Yes	No
Lipopolysaccharide (endotoxin)	No	Yes



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- Peptidoglycan in gram-positive bacteria is much thicker than in gram-negative bacteria.
- Only gram-negative bacteria have an outer membrane containing endotoxin (lipopolysaccharide [LPS]).
- Gram-negative bacteria have a periplasmic space where β-lactamases are found. (the use of inhibitors). (Gram-positive also have β-lactamases)
- Several important gram-positive bacteria, such as staphylococci and streptococci, have teichoic acids.

- β-lactamase inhibitors: Clavulanic acid, sulbactam, and tazobactam

Components of Bacterial Cells

Structure	Composition	Distribution[a]		
		Gram-Negative Cell	Gram-Positive Cell	Mollicutes (Mycoplasmas)
Envelope				
Capsule (slime layer)	Polysaccharide or polypeptide	+ or −	+ or −	−
Wall		+	+	−
Outer membrane	Proteins, phospholipids, and lipopolysaccharide	+	−	−
Peptidoglycan layer	Murein (+ teichoate in Gram-positive cells)	+	+	−
Periplasm	Proteins and oligosaccharides in solution	+	−	−
Cell membrane	Proteins, phospholipids	+	+	+

The peptidoglycan layer is much thicker in gram-positive than in gram-negative bacteria. Many gram-positive bacteria also have fibers of teichoic acid that protrude outside the peptidoglycan and gives rigidity to cell wall, whereas gram-negative bacteria do not have teichoic acids.

- Gram-negative bacteria have a complex outer layer consisting of lipopolysaccharide, lipoprotein, and phospholipid.

- Lying between the outer-membrane layer and the cytoplasmic membrane in gram-negative bacteria is the periplasmic space, which is the site, in some species, of enzymes called β-lactamases that degrade penicillins and other β-lactam drugs.

Other important properties of the cell wall:

- In gram-negative bacteria, it contains endotoxin, a lipopolysaccharide
- Its polysaccharides and proteins are antigens that are useful in laboratory identification.
- Its porin proteins play a role in facilitating the passage of small hydrophilic molecules into the cell.
- Porin proteins in the outer membrane of gram-negative bacteria act as a channel to allow the entry of essential substances such as sugars, amino acids, vitamines, and metals as well as many antimicrobial drugs such as penicillins.

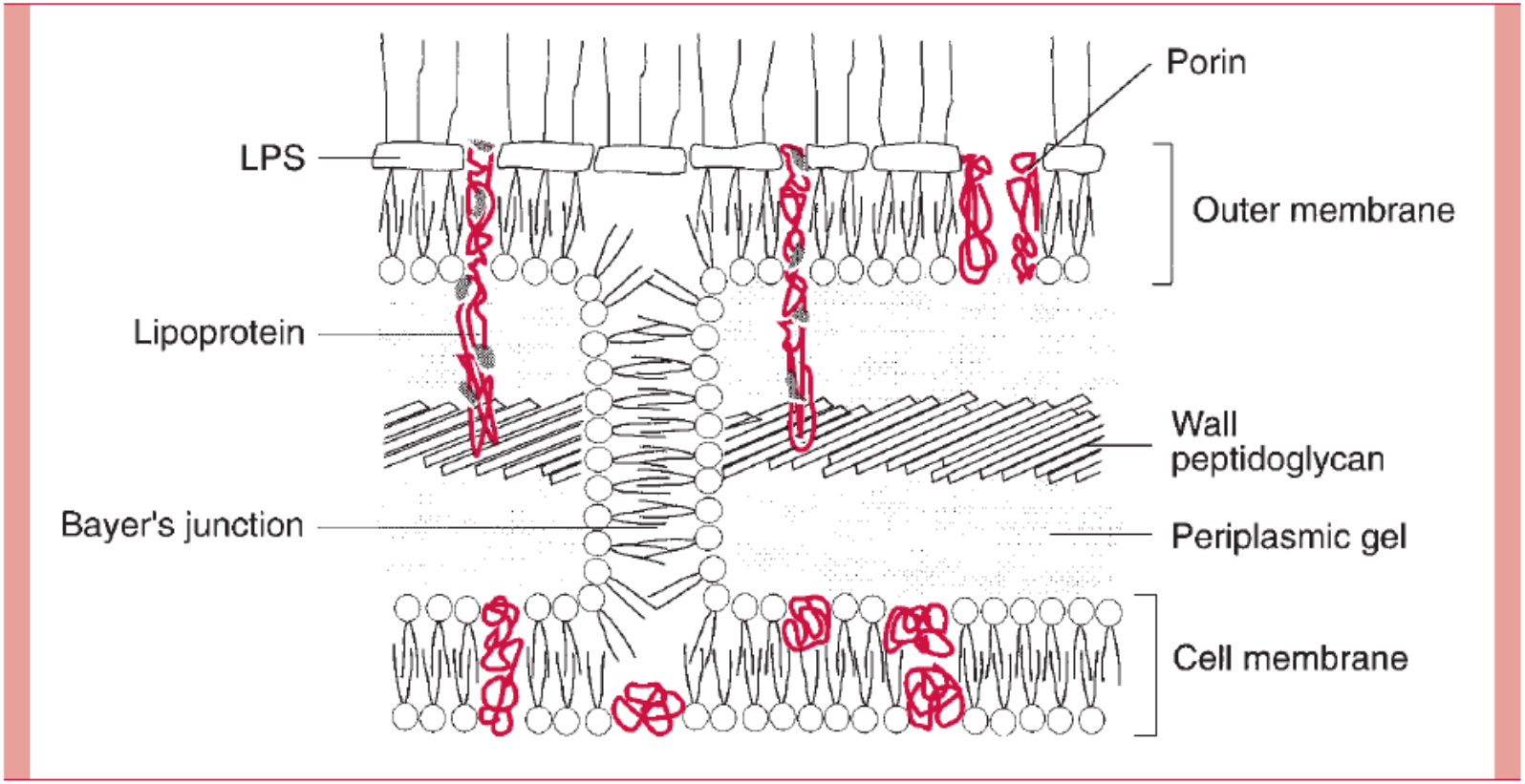


FIGURE 2–9

Schematic representation of wall of Gram-negative bacteria. LPS, lipopolysaccharide with endotoxic properties.

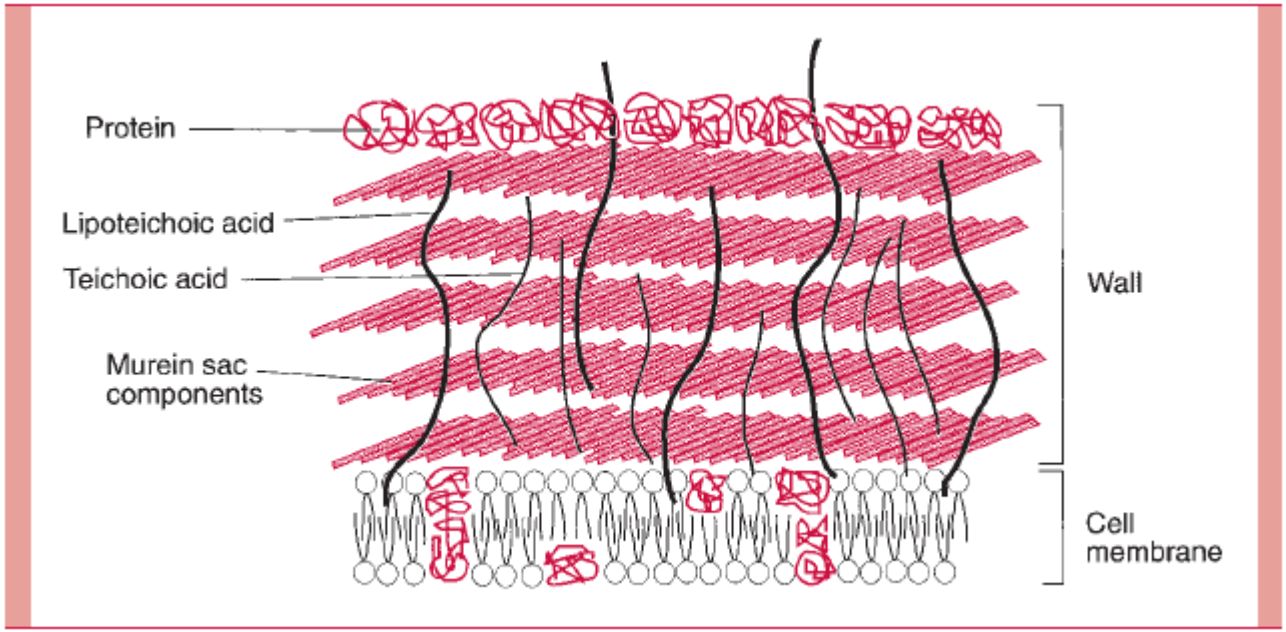


FIGURE 2–6
Schematic representation of the wall of Gram-positive bacteria.

Three important components of the cell wall:

1- Peptidoglycan, 2- Lipopolysaccharide, 3- Teichoic acid.

Peptidoglycan is found *only* in bacterial cell walls. It provides rigid support for the cell, is important in maintaining the characteristic shape of the cell, and allows the cell to withstand media of low osmotic pressure, such as water.

Peptidoglycan is derived from the peptides and the sugars (glycan) that make up the molecule.

Synonyms for peptidoglycan are murein and mucopeptide.

In Latin word murus, meaning "wall." /// murein, the bacterial cell-wall material

Lipopolysaccharide: It has 3 parts:

1- Lipid A

- **Anchors LPS in the outer membrane.**
- **Responsible for the endotoxin activity.**
- **Triggers strong innate immune responses.**

2- Core polysaccharide

- **Connects Lipid A to the O-antigen.**

3- O-antigen (O-polysaccharide)

- **Outward of the bacterial surface.**
- **Highly variable between bacterial strains.**
- **Important for serotyping and immune recognition (Somatic antigen).**

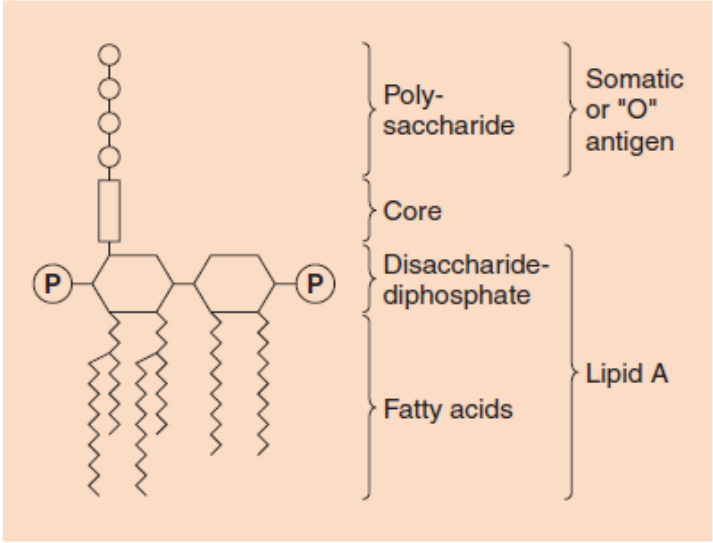
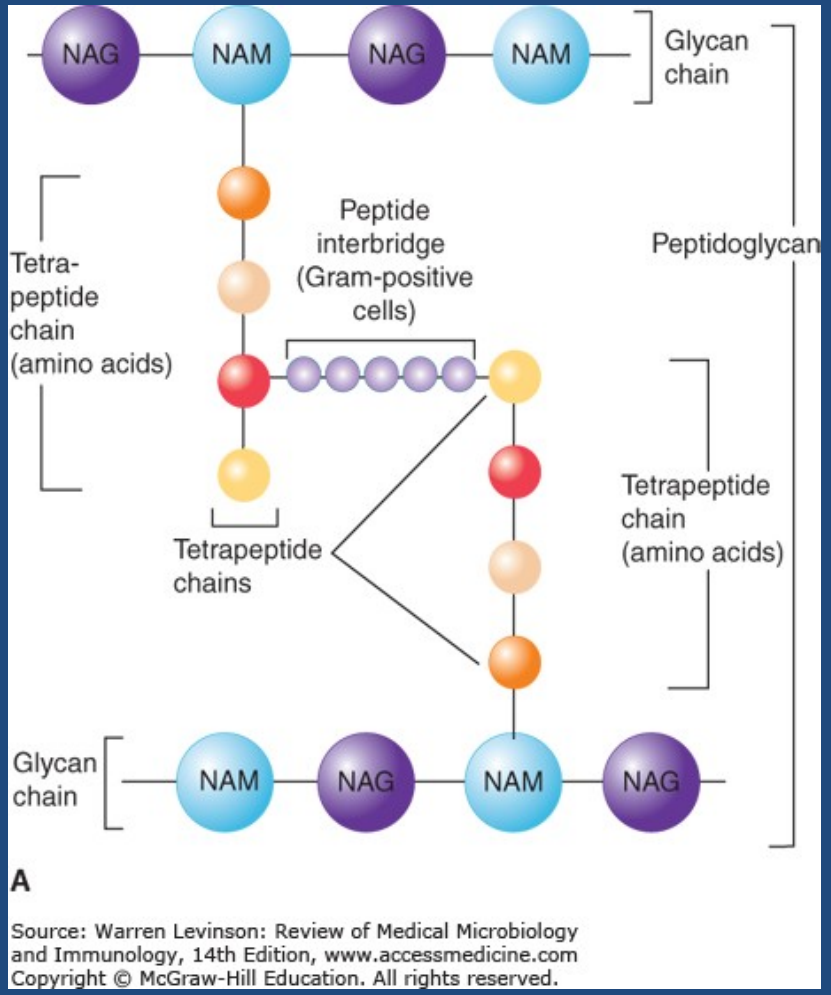


FIGURE 2–6 Endotoxin (lipopolysaccharide [LPS]) structure. The O-antigen polysaccharide is exposed on the exterior of the cell, whereas the lipid A faces the interior. (Reproduced with permission from Brooks GF et al. *Medical Microbiology.* 19th ed. Originally published by Appleton & Lange. Copyright 1991, McGraw-Hill.)

Peptidoglycan structure: is composed of a glycan chain (NAM and NAG), a tetrapeptide chain, and a cross-link (peptide interbridge).

- NAG, *N*-acetylglucosamine.

- NAM, *N*-acetylmuramic acid.

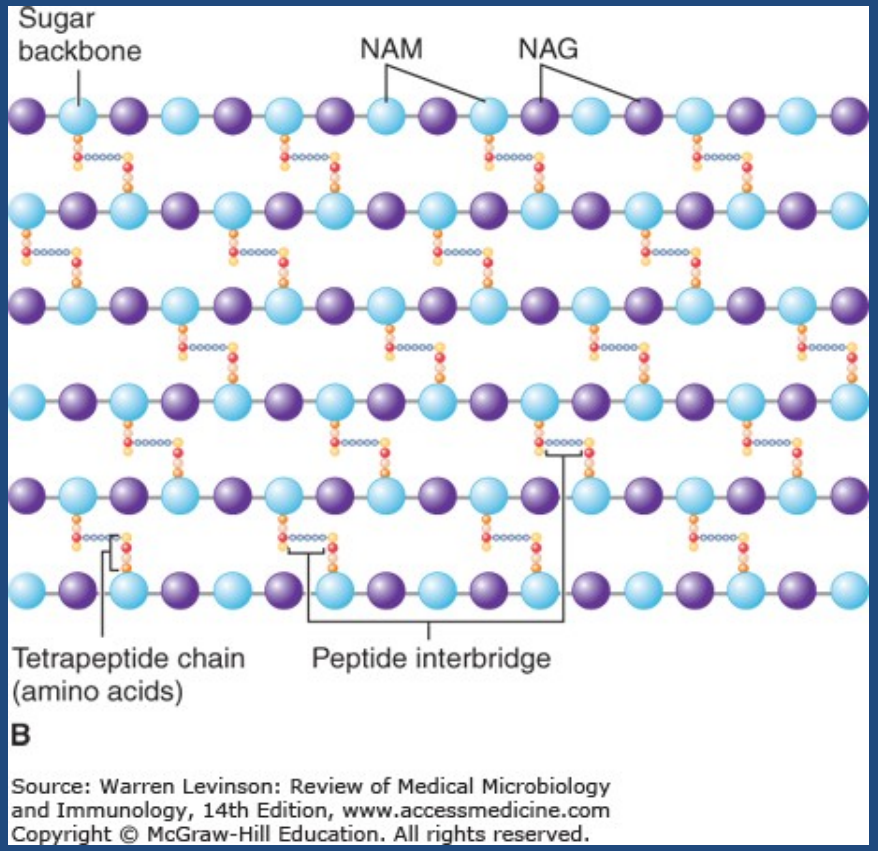


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In the cell wall, the peptidoglycan forms a multilayered, three-dimensional structure.

NAG: *N*-acetylglucosamine.

NAM: *N*-acetylmuramic acid.



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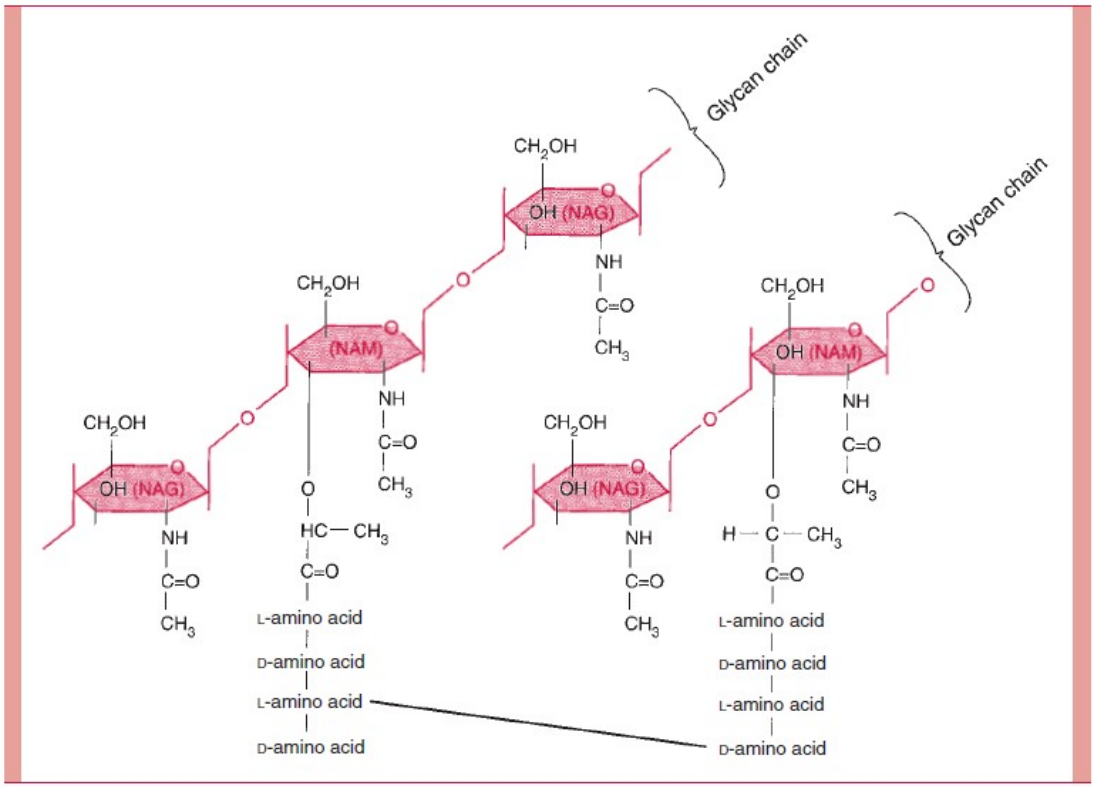


FIGURE 2–7

Schematic representation of the peptidoglycan murein. NAG, *N*-acetylglucosamine; NAM, *N*-acetylmuramic acid.

D-amino acids Versus L-amino acids !!!!

Teichoic Acid:

- Teichoic acids are fibers located in the outer layer of the gram-positive cell wall.
- They are composed of polymers of either glycerol phosphate or ribitol phosphate.
- Some polymers of glycerol teichoic acid penetrate the peptidoglycan layer and are covalently linked to the lipid in the cytoplasmic membrane, in which case they are called lipoteichoic acid; others anchor to the muramic acid of the peptidoglycan.

The medical importance of teichoic acids lies in their ability to induce inflammation and septic shock (sepsis) when caused by certain gram-positive bacteria, they activate the same pathways as does endotoxin (LPS) in gram-negative bacteria.
- Teichoic acids also mediate the attachment of staphylococci to mucosal cells.

GRAM STAIN

- Developed in 1884 by the Danish physician Christian Gram.

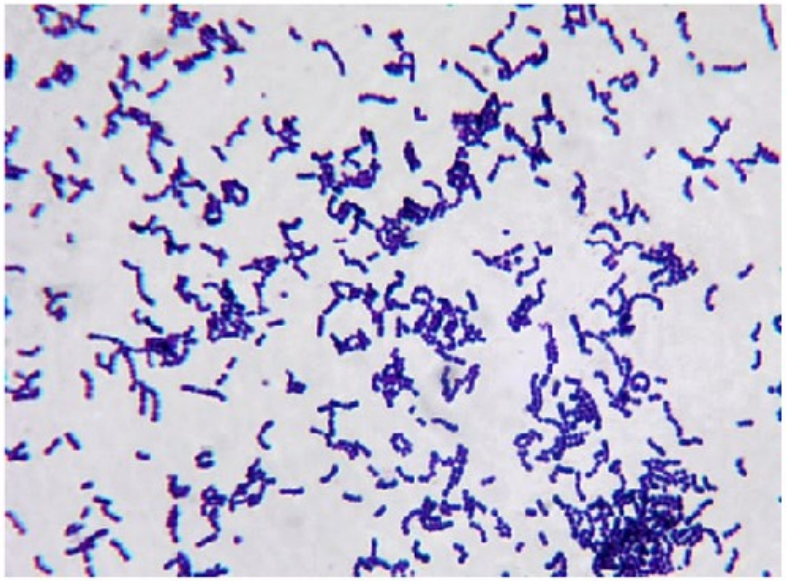
- It separates most bacteria into two groups: the gram-positive bacteria, which stain blue, and the gram-negative bacteria, which stain red.

-The Gram stain involves the following four-step :

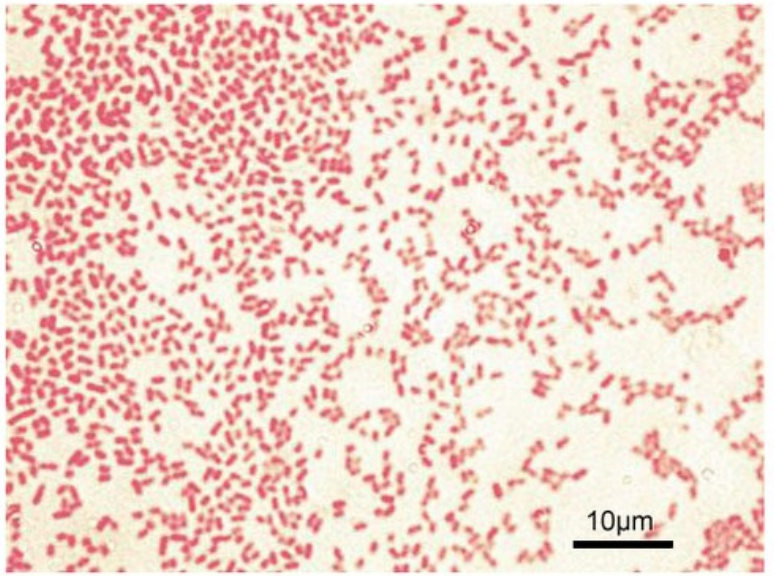
1- The crystal violet dye stains all cells blue/purple.
2- The iodine solution (mordant) is added to form a crystal violet–iodine complex.

3- Decolorization by acetone or ethanol, extracts the blue dye complex from the lipid-rich, thin-walled gram-negative bacteria to a greater degree than from the lipid-poor, thick-walled gram-positive bacteria.

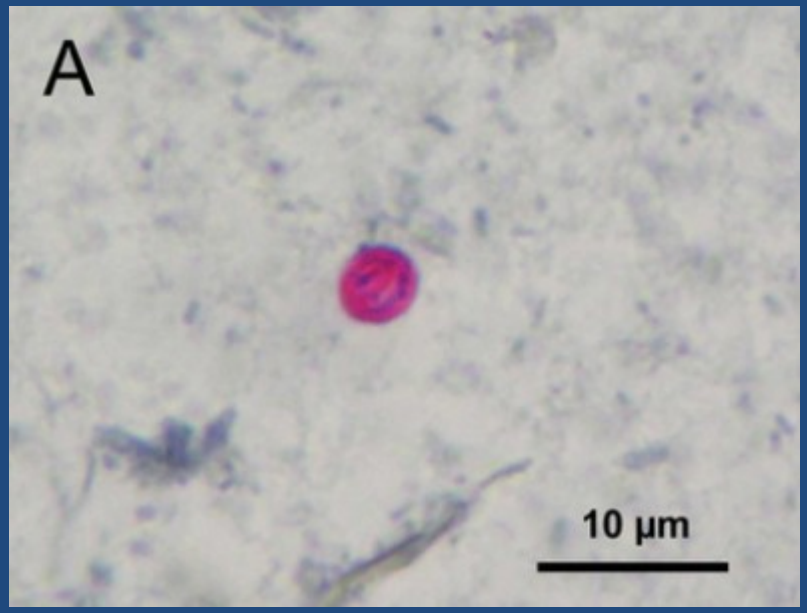
4- The red dye safranin stains the decolorized gram-negative cells red/pink; the gram-positive bacteria remain blue.

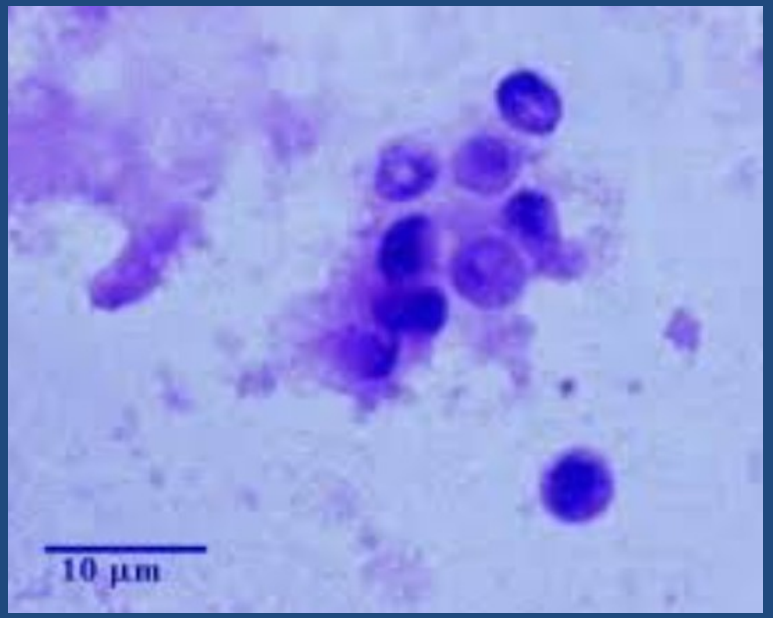


Gram Positive Bacteria



Gram Negative Bacteria


A
10 µm


10 µm

The Gram stain is useful in two ways:

1- In the identification of many bacteria.

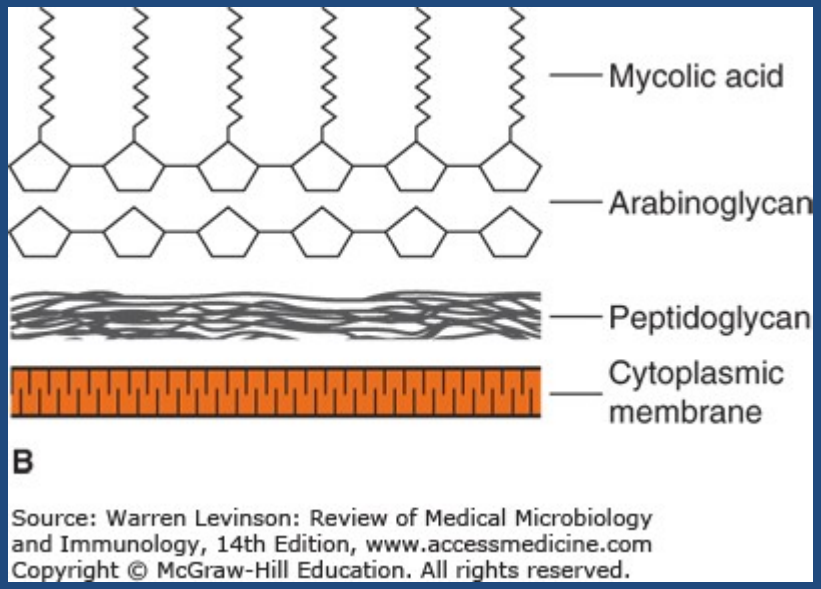
2- In influencing the choice of antibiotic because, in general, gram-positive bacteria are more susceptible to penicillin G than are gram-negative bacteria (empirical treatment).

Medically Important Bacteria That Cannot Be Seen in the Gram Stain

Name	Reason	Alternative Microscopic Approach
Mycobacteria, including *M. tuberculosis*	Too much lipid in cell wall so dye cannot penetrate	Acid-fast stain
Treponema pallidum	Too thin to see	Dark-field microscopy or fluorescent antibody
Mycoplasma pneumoniae	No cell wall; very small	None
Legionella pneumophila	Poor uptake of red counterstain	Prolong time of counterstain
Chlamydiae, including *C. trachomatis*	Intracellular; very small	Inclusion bodies in cytoplasm
Rickettsiae	Intracellular; very small	Giemsa or other tissue stains

Cell Walls of Acid-Fast Bacteria

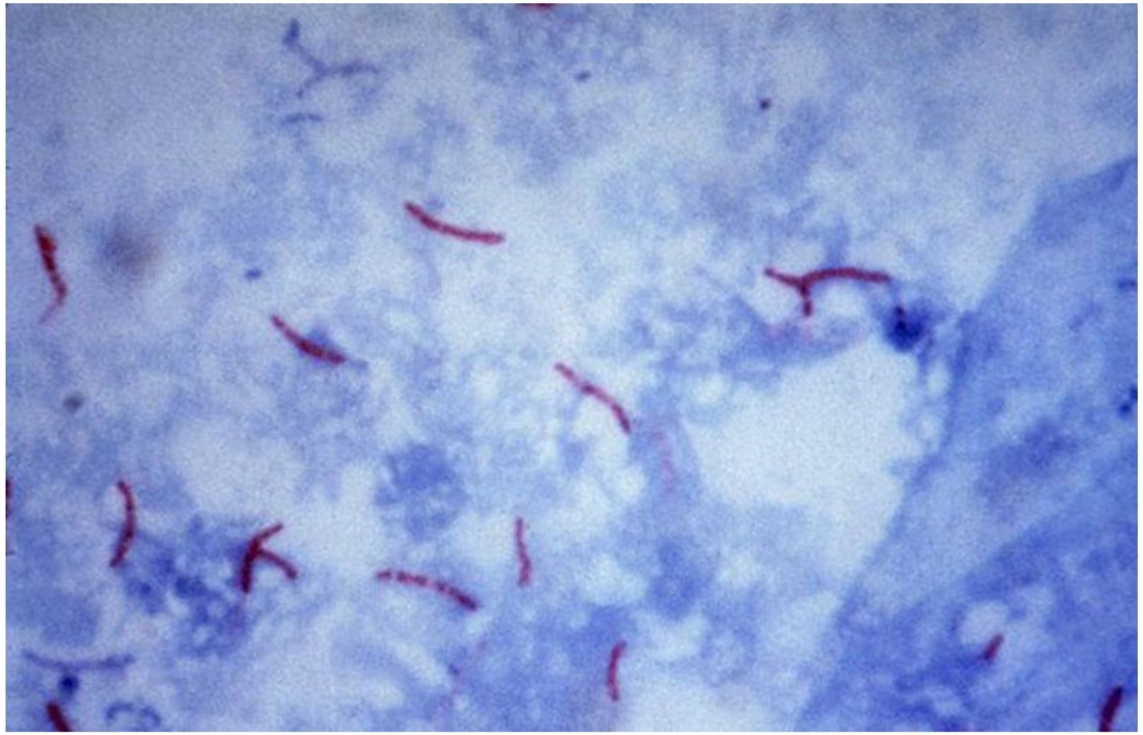
- Mycobacteria (e.g., *Mycobacterium tuberculosis*) have an unusual cell wall, resulting in their inability to be Gram-stained.
- These bacteria are said to be acid-fast because they resist decolorization with acid–alcohol after being stained with carbolfuchsin.
- Acid-fast bacteria: Red color using carbolfuchsin
- Non–acid-fast: Blue color (using methylene blue counter stain)
- This property is related to the high concentration of lipids, called mycolic acids, in the cell wall of mycobacteria.



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Cell wall of *Mycobacterium tuberculosis:*

The layers of mycolic acid (long chain fatty acids) and arabinoglycan (polysaccharides) are present in members of the genus *Mycobacterium* but not in most other genera of bacteria.



Mycobacterium tuberculosis **visualization using the Ziehl–Neelsen stain.**

Lysozyme:

An enzyme present in human tears, mucus, and saliva, can cleave the peptidoglycan backbone by breaking its glycosyl bonds ***(hydrolysis of 1,4-beta-linkages between N-acetylmuramic acid and N-acetyl-D-glucosamine).***

- Lysozyme-treated bacteria may swell and rupture as a result of the entry of water into the cells.

- Lysozyme-treated bacteria can survive in isotonic solution, as spherical forms, called protoplasts, surrounded only by a cytoplasmic membrane.

Bacterial Structures

Structure	Chemical Composition	Function
Essential components		
Cell wall		
Peptidoglycan	Glycan (sugar) backbone with peptide side chains that are cross-linked	Gives rigid support, protects against osmotic pressure, is the site of action of penicillins and cephalosporins, and is degraded by lysozyme
Outer membrane of gram-negative bacteria	Lipid A	Toxic component of endotoxin
	Polysaccharide	Major surface antigen used frequently in laboratory diagnosis
Surface fibers of gram-positive bacteria	Teichoic acid	Major surface antigen but rarely used in laboratory diagnosis

Essential components

Plasma membrane	Lipoprotein bilayer without sterols	Site of oxidative an transport enzymes
Ribosome	RNA and protein in 50S and 30S subunits	Protein synthesis; site of action of aminoglycosides, erythromycin, tetracyclines, and chloramphenicol
Nucleoid	DNA	Genetic material
Mesosome	Invagination of plasma membrane	Participates in cell division and secretion
Periplasm	Space between plasma membrane	Contains many hydrolytic enzymes

Nonessential components		
Capsule	Polysaccharide[1]	Protects against phagocytosis
Pilus or fimbria	Glycoprotein	Two types: (1) mediates attachment to cell surfaces; (2) sex pilus mediates attachment of two bacteria during conjugation
Flagellum	Protein	Motility
Spore	Keratinlike coat, dipicolinic acid	Provides resistance to dehydration, heat, and chemicals
Plasmid	DNA	Contains a variety of genes for antibiotic resistance and toxins
Granule	Glycogen, lipids, polyphosphates	Site of nutrients in cytoplasm
Glycocalyx	Polysaccharide	Mediates adherence to surfaces

Cytoplasmic Membrane

- Composed of a phospholipid bilayer similar in microscopic appearance to that in eukaryotic cells.
- Eukaryotic membranes contain sterols, whereas prokaryotes generally do not. (Erog sterols).
- The only prokaryotes that have sterols in their membranes are members of the genus *Mycoplasma*.

- The membrane has four important functions:

1- active transport of molecules into the cell.

2- energy generation by oxidative phosphorylation.

3- synthesis of precursors of the cell wall.

4-secretion of enzymes and toxins.

Ribosomes: Are the site of protein synthesis. Bacterial ribosomes are 70S in size, with 50S and 30S subunits, whereas eukaryotic ribosomes are 80S in size, with 60S and 40S subunits. The differences in both the ribosomal RNAs and proteins constitute the basis of the selective action of several antibiotics that inhibit bacterial, but not human, protein synthesis.

Granules (inclusion bodies): Are Different types, that serve as storage areas for nutrients and they can be stained with certain dyes. (Found in *Corynebacterium*, *Yersinia*, and *Mycobacterium*),,,,(*Glycan granules in E. coli*).

Structures Outside the Cell Wall

1- Capsule

A gelatinous layer covering the entire bacterium. It is composed of polysaccharide (some bacteria have a polypeptide capsul), except in the anthrax bacillus, which has a capsule of polymerized d-glutamic acid.

- The sugar components of the polysaccharide vary from one species of bacteria to another and frequently determine the serologic type (serotype) within a species.

- There are 84 different serotypes of *Streptococcus pneumoniae*, which are distinguished by the antigenic differences of the sugars in the polysaccharide capsule. (*Salmonella* /2,600, *Vibrio* /200)

Importance of the capsule :

1- It is a determinant of virulence of many bacteria since it limits the ability of phagocytes to engulf the bacteria. Negative charges on the capsular polysaccharide repel the negatively charged cell membrane of the neutrophil and prevent it from ingesting the bacteria.

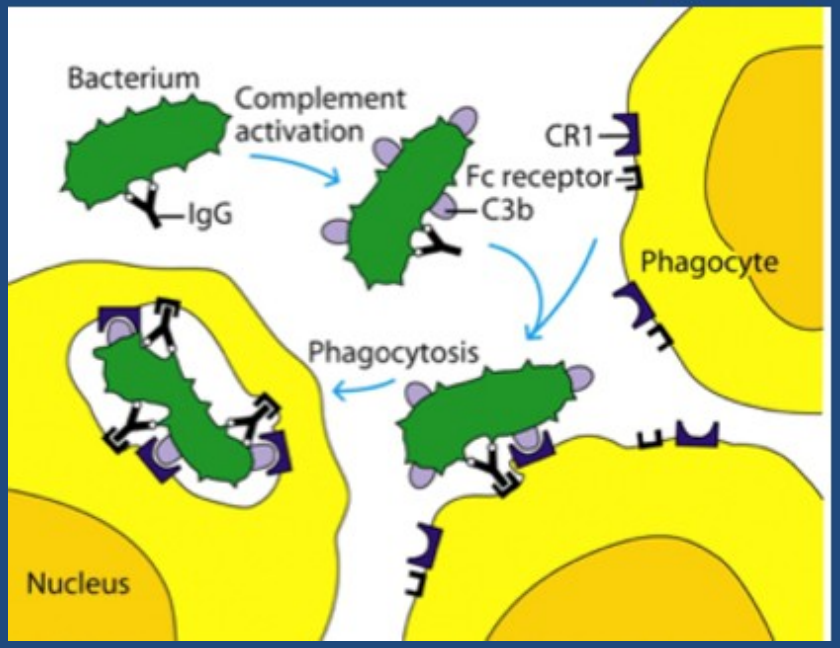
2- Specific identification using antiserum against the capsular polysaccharide. (Quellung reaction / capsule swelling test!!).

3- Capsular polysaccharides are used as the antigens in certain vaccines because they are capable of eliciting protective antibodies.

4- The capsule may play a role in the adherence of bacteria to human tissues, which is an important initial step in causing infection.

- Capsules are antiphagocytic; that is, they limit the ability of neutrophils to engulf the bacteria.

- Antibodies against the capsule or complement proteins act help in phagocytic cells to engulf bacteria.



Opsonization

- Opsonization is the process by which antibodies enhance the phagocytosis of bacteria.

2- Flagella:

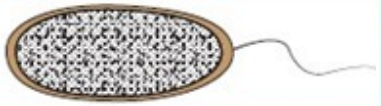
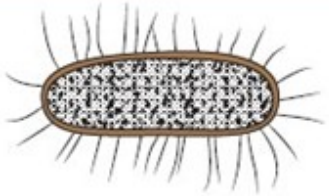
- Are long, whiplike appendages that move the bacteria toward nutrients and other attractants, a process called chemotaxis.

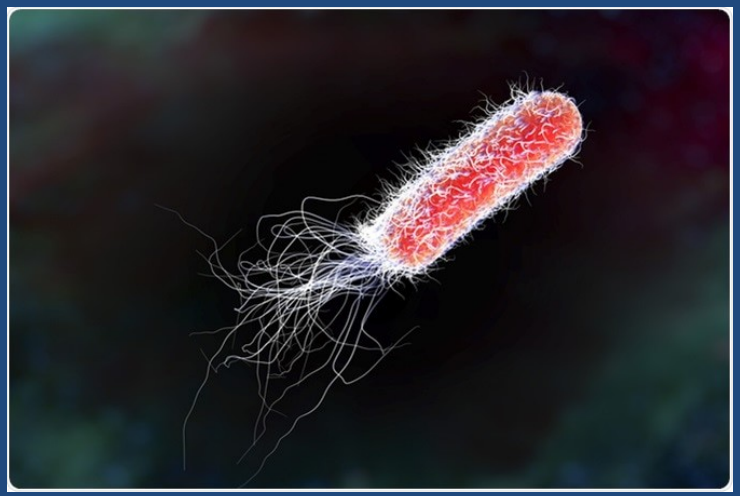
- Falgella is composed of many subunits of a single protein, flagellin.
- Flagella need energy for movement (ATP).

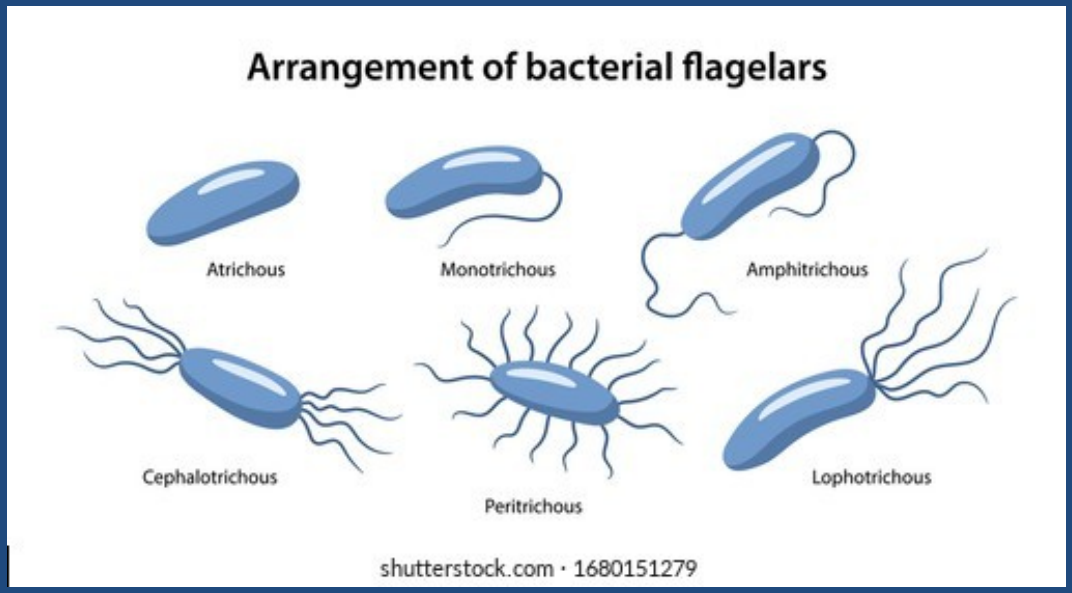
- Flagellated bacteria have a characteristic number and location of flagella.

- Many rods do have flagella, but most cocci do not have and are therefore nonmotile.

- Spirochetes have axial flagella/filament

Structure	Example
	*Vibrio cholera*
	*Pseudomonas fluorescens*
	*Aquaspirillum serpens*
	*Salmonella typhi*





Flagella are medically important for two reasons:

- Movement:

- Some species of motile bacteria (e.g., *E. coli* and *Proteus* species) are common causes of urinary tract infections. (propelling the bacteria up the urethra into the bladder)

- Diagnosis:

- Some species of bacteria (e.g., *Salmonella* species) are identified in the clinical laboratory by the use of specific antibodies against flagellar proteins.

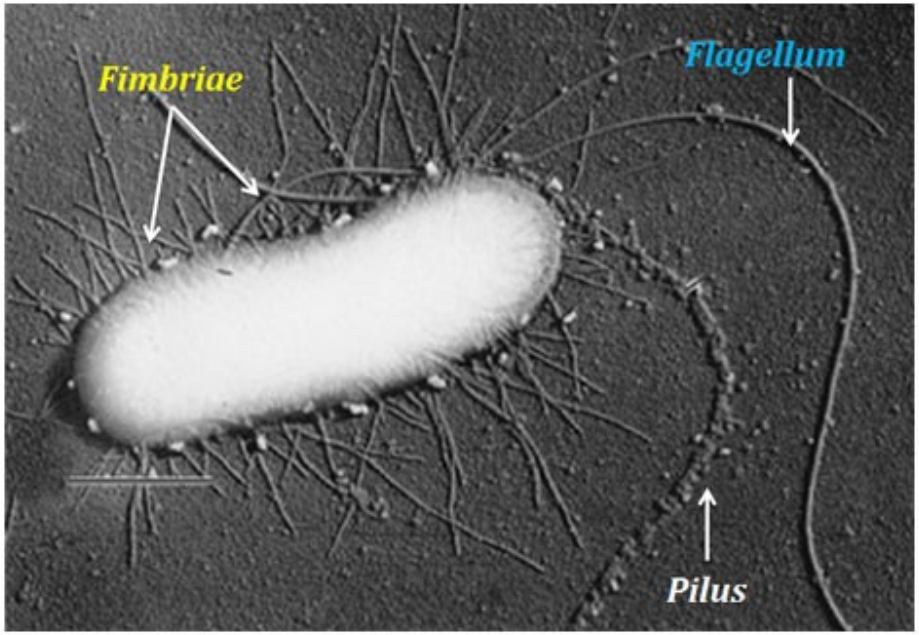
3- Pili (Fimbriae): / (in Latin: threads)

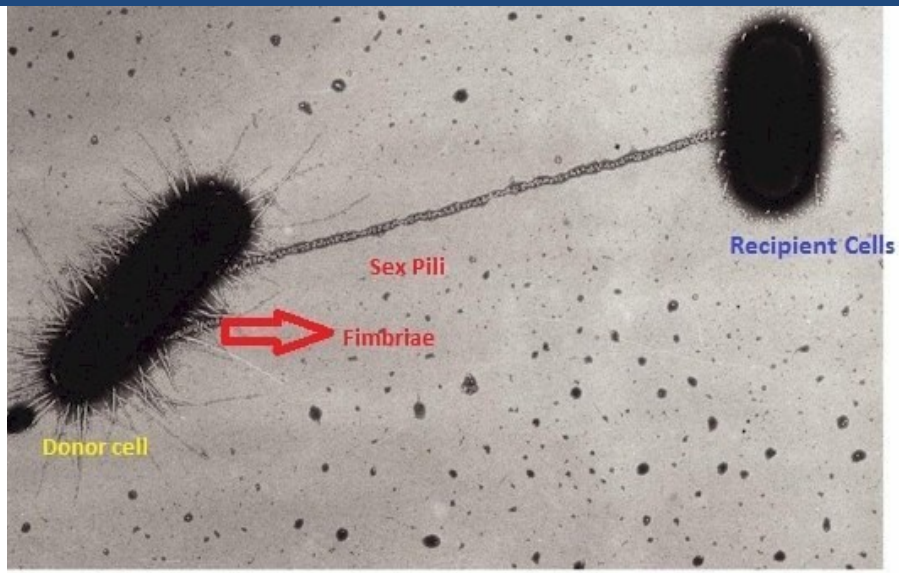
- Are hair-like filaments that extend from the cell surface. They are shorter and straighter than flagella and are composed of subunits of pilin, a protein arranged in helical strands. They are found mainly on gram-negative organisms. (Some Gram-positive bacteria have pilli)

- Pili are often important virulence factors.
- They mediate the attachment of bacteria to specific receptors on the human cell surface, which is a necessary step in the initiation of infection for some organisms.

- Mutants of *Neisseria gonorrhoeae* that do not form pili are nonpathogens.

- A specialized kind of pilus, the sex pilus, is used in conjugation.





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TEM

1 µm

4- Glycocalyx //(two main forms: Capsule and Slime Layer):

The glycocalyx is a polysaccharide coating that is secreted by many bacteria. It covers surfaces like a film and allows the bacteria to adhere firmly to various structures (e.g., skin, heart valves, prosthetic joints, and catheters).

-The glycocalyx is an important component of biofilms.

- Medical importance:
- Glycocalyx-producing strains of P. aeruginosa that cause respiratory tract infections in cystic fibrosis patients.
- Glycocalyx-producing strains of Staphylococcus epidermidis and viridans streptococci that cause endocarditis.

- The glycocalyx also mediates adherence of certain bacteria, such as *Streptococcus mutans*, to the surface of teeth. This plays an important role in the formation of plaque, the precursor of dental caries (تسوس الأسنان).

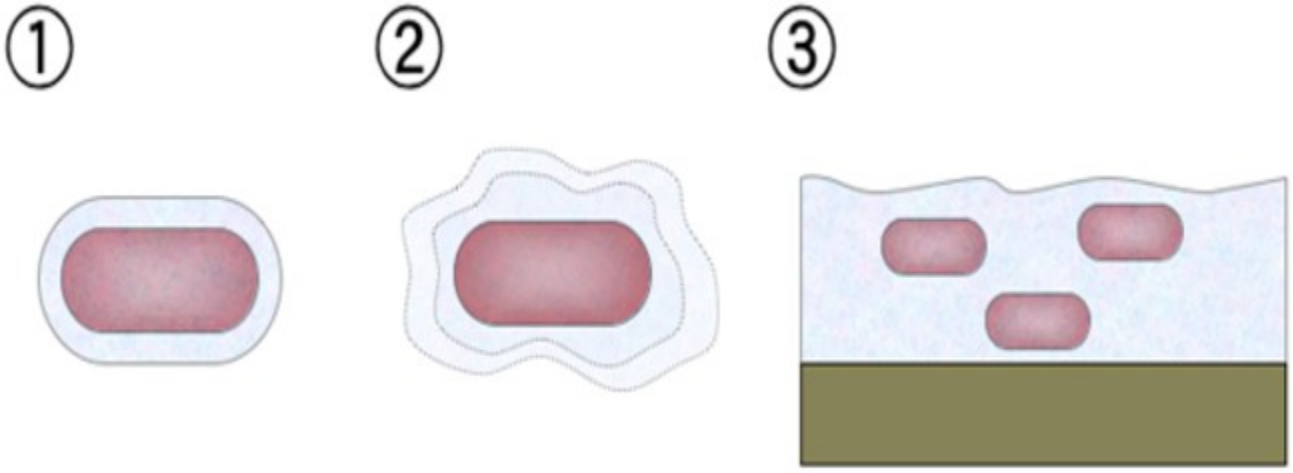


Diagram of bacterial mucoid-like structures:
1. capsule; 2. slime layer; 3. biofilm

Bacterial Spores

- Highly resistant formed in response to hard conditions.

Mainly in: gram-positive rods:

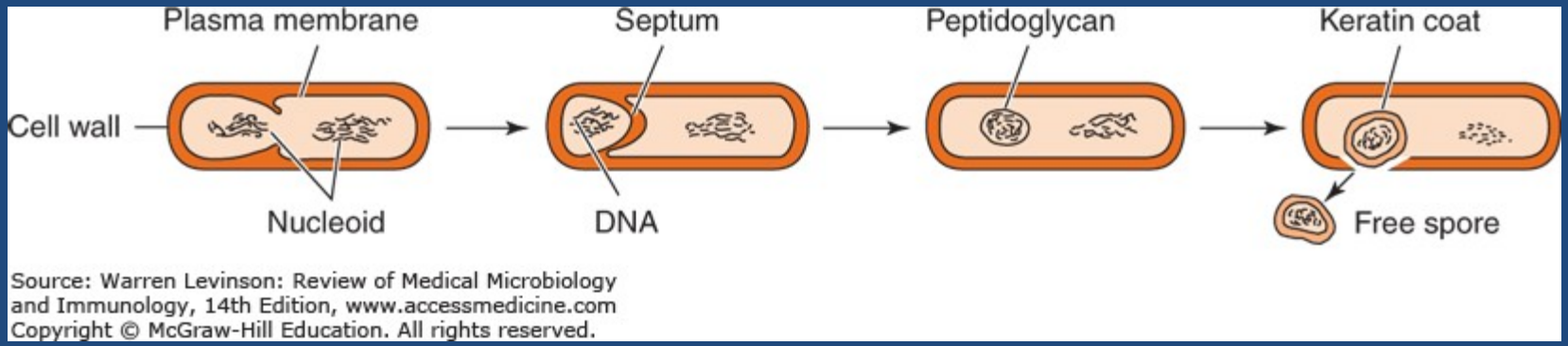
- genus *Bacillus*, (including the agent of anthrax / ***Bacillus anthracis***).
- genus *Clostridium*, (including the agents of tetanus).

- Spore formation (sporulation) occurs when nutrients are depleted.

- The spore forms inside the cell with a thick, keratin-like coat that gives **resistance of the spore to heat, dehydration, radiation, and chemicals.**

- The resistance is mediated by dipicolinic acid (pyridine-2,6-dicarboxylic acid), a calcium ion chelator found only in spores.

- Spore has no metabolic activity and can remain dormant for many years.



Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
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Bacterial spores. The spore contains the entire DNA genome of the bacterium surrounded by a thick, resistant coat.

As a result of their resistance to heat, sterilization cannot be achieved by boiling. Steam heating under pressure (autoclaving) at 121°C, for at least 15 minutes, is required to ensure the sterility of products for medical use.

Good conditions: spores germinate to vegetative cells

Important Features of Spores

Highly resistant to heating

spores are not killed by boiling (100°C).

Highly resistant to many chemicals, including most disinfectants, due to the thick, keratin like coat of the spore.

They can survive for many years, especially in the soil.

No measurable metabolic activity.

antibiotics.

Classification of Medically Important Bacteria

Based on morphologic and biochemical characteristics.

Only medically important bacteria are discussed.

Gram-negative rods, are divided into three categories:

- Respiratory organisms,
- Zoonotic organisms,
- Enteric and related organisms.

The initial criterion used in the classification is the nature of the cell wall : - rigid - flexible - absent.

- Bacteria with rigid, thick walls can be subdivided into:

1- free-living bacteria, that can grow on laboratory medium without human or other animal cells,

2-non–free-living bacteria, which are obligate intracellular

- The free-living organisms are further subdivided according to shape (cocci and rods) and gram-satin. Also these are divided into their oxygen requirements and spore-forming abilities.

- Bacteria with flexible, thin walls (the spirochetes) and those without cell walls (the mycoplasmas) form separate units.

- Based on this many bacteria can be readily classified into separate genus and species. (rRNA / DNA based classification).

Classification of Medically Important Bacteria (Table 5-1/page 25)

Characteristics	Genus	Representative Diseases
I. Rigid, thick-walled cells		
A. Free-living (extracellular bacteria)		
1. Gram-positive		
a. Cocci	*Streptococcus*	Pneumonia, pharyngitis, cellulitis
	Staphylococcus	Abscess of skin and other organs
b. Spore-forming rods		
(1) Aerobic	*Bacillus*	Anthrax
(2) Anaerobic	*Clostridium*	Tetanus, gas gangrene, botulism
c. Non–spore-forming rods		
(1) Nonfilamentous	*Corynebacterium*	Diphtheria
	Listeria	Meningitis
(2) Filamentous	*Actinomyces*	Actinomycosis
	Nocardia	Nocardiosis
2. Gram-negative		

2. Gram-negative		
a. Cocci	*Neisseria*	Gonorrhea, meningitis
b. Rods		
(1) Facultative		
(a) Straight		
(i) Respiratory organisms	*Haemophilus*	Meningitis
	Bordetella	Whooping cough
	Legionella	Pneumonia
(ii) Zoonotic organisms	*Brucella*	Brucellosis
	Francisella	Tularemia
	Pasteurella	Cellulitis
	Yersinia	Plague
(iii) Enteric and related organisms	*Escherichia*	Urinary tract infection, diarrhea
	Enterobacter	Urinary tract infection
	Serratia	Pneumonia
	Klebsiella	Pneumonia, urinary tract infection
	Salmonella	Enterocolitis, typhoid fever
	Shigella	Enterocolitis
	Proteus	Urinary tract infection

(b) Curved	*Campylobacter*	Enterocolitis
	Helicobacter	Gastritis, peptic ulcer
	Vibrio	Cholera
(2) Aerobic	*Pseudomonas*	Pneumonia, urinary tract infection
(3) Anaerobic	*Bacteroides*	Peritonitis
3. Acid-fast	*Mycobacterium*	Tuberculosis, leprosy
B. Non–free-living (obligate intracellular parasites)	*Rickettsia*	Rocky Mountain spotted fever, typhus, Q fever
	Chlamydia	Urethritis, trachoma, psittacosis
II. Flexible, thin-walled cells (spirochetes)		Syphilis
	Borrelia	Lyme disease
	Leptospira	Leptospirosis
III. Wall-less cells	*Mycoplasma*	Pneumonia

Characteristics	Genus	Representative Diseases
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	Listeria	Meningitis
(2) Filamentous	*Actinomyces*	Actinomycosis
	Nocardia	Nocardiosis

Free-living / Non free-living

2. Gram-negative		
a. Cocci	*Neisseria*	Gonorrhea, meningitis
b. Rods		
(1) Facultative		
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(I) Respiratory organisms	*Haemophilus*	Meningitis
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	Chlamydia	Urethritis, trachoma, psittacosis
II. Flexible, thin-walled cells (spirochetes)	*Treponema*	Syphilis
	Borrelia	Lyme disease
	Leptospira	Leptospirosis
III. Wall-less cells	*Mycoplasma*	Pneumonia